

THARINDU JAYASANKHA

SOFTWARE ENGINEER

Kalutara, Sri Lanka • (+94) 71 206 70 24 • tharindujayasankha@gmail.com
linkedin.com/in/tharindu-jayasankha • github.com/tharindu1999 • tharindu1999.github.io/portfolio

PROFILE SUMMARY

Full stack software engineer with over three years of experience building and scaling enterprise applications end to end, from React micro frontends to .NET microservices running on Azure and Kubernetes. Increasingly focused on applied AI, with hands on delivery of production RAG chatbots, LLM integrations, and AI assisted engineering workflows.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, Python

Backend Development: ASP.NET Core (Web API, Identity), Entity Framework Core, Node.js, REST, GraphQL

Frontend Development: React.js, Tailwind CSS, Ant Design, Nx Workspace (micro frontend architecture)

Databases & Messaging: PostgreSQL, SQL Server, MongoDB, Redis, RabbitMQ

Cloud & DevOps: Microsoft Azure, Docker, Kubernetes, Azure DevOps, CI/CD pipelines, ELK Stack, Grafana

AI & Machine Learning: RAG pipelines, LangChain, OpenAI and Gemini APIs, Azure Cognitive Services, vector databases (Chroma, pgvector), agentic coding (Cursor, Codex, Claude), PyTorch, TensorFlow, Hugging Face Transformers, OpenCV

Parsing & Workflow Engines: Nearley.js, Moo, domain specific languages (DSLs), rule engines

CAREER EXPERIENCE

Software Engineer

2024 – Present

Millennium IT ESP

- Migrated the platform's authentication and authorization model from role based (RBAC) to attribute based access control (ABAC), enabling finer grained permissions across services.
- Configured ASP.NET Identity Server 4 for secure token management and centralized authentication.
- Deployed and operated containerized microservices on Docker and Kubernetes in Azure, maintaining scalability and high availability in production.
- Implemented asynchronous inter service communication with RabbitMQ, alongside caching and session management with Redis.
- Designed and tuned relational (SQL Server, PostgreSQL) and NoSQL (MongoDB) databases for performance and reliability.
- Built and optimized production chatbot solutions on Retrieval Augmented Generation (RAG) pipelines using Python, LangChain, Node.js, and vector databases.
- Led research and development on cloud native AI, applying Azure Cognitive Services and multiple LLMs to deliver real time speech to text processing with speaker diarization.
- Automated image validation workflows in Python, improving the accuracy and reliability of data processing pipelines.

Associate Software Engineer

2023 – 2024

Millennium IT ESP

- Built responsive, scalable front end applications with React.js, TypeScript, and Nx Workspace as part of a micro frontend architecture.
- Designed accessible, cross device UI components using Tailwind CSS and Ant Design.
- Developed RESTful Web APIs in ASP.NET Core, using Entity Framework Core and LINQ for data access and business logic.
- Integrated GraphQL APIs to streamline client–server communication in enterprise applications.
- Engineered a tokenizer and modular rule engine in Node.js (Nearley.js, Moo) capable of parsing complex domain specific languages and validating high volume data streams at low latency.

Intern Software Engineer

2022 – 2023

Millennium IT ESP

- Contributed to a React.js spreadsheet style calculation and reporting system, implementing dynamic formulas, data validation, and custom report generation.
- Wrote unit and integration tests to safeguard software reliability and maintainability.
- Worked in a Scrum team using Git and Azure DevOps pipelines, taking part in sprint planning, daily stand ups, and code reviews.

EDUCATIONAL QUALIFICATIONS

Master of Science in Artificial Intelligence (Expected 2026)

University of Moratuwa

Bachelor of Science (Honours) in Computer Science – Second Class Upper Division (2020 – 2024)

University of Westminster, UK (Informatics Institute of Technology)

RESEARCH AREA

Gemstone Identification and Authentication Using Vision Transformers

Developed and implemented Vision Transformer (ViT) models for image classification, applied to the real world problem of identifying and authenticating gemstones without reliance on expert knowledge or specialized tools. Leveraged self attention mechanisms to capture global image features, outperforming traditional CNN approaches, and built an image preprocessing pipeline to improve input quality and classification reliability. Enhanced model accuracy through fine tuning, data augmentation, and hyperparameter optimization, achieving 75.87% testing accuracy with an F1 score of 0.72 (precision 0.76, recall 0.73).

REFERENCES

Available upon request.